

MAE 6291

Internet of Things for Engineers

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Week 4 [02/11/2026]

- Setting up the Edge Lab
- IoT Architecture and Ecosystem
- Layers in IoT systems - 3 level layer model
- The Linux Terminal
- Setting up a Python virtual environment [venv] - Sandboxing your code
- Bash script to run RPi-python code
- Blinking LEDs on boot [Graded lab activity]

`git clone https://github.com/gwu-mae6291-iot/spring2026_codes.git`



School of Engineering
& Applied Science

Spring 2026

THE GEORGE WASHINGTON UNIVERSITY

Photo: Kartik Bulusu

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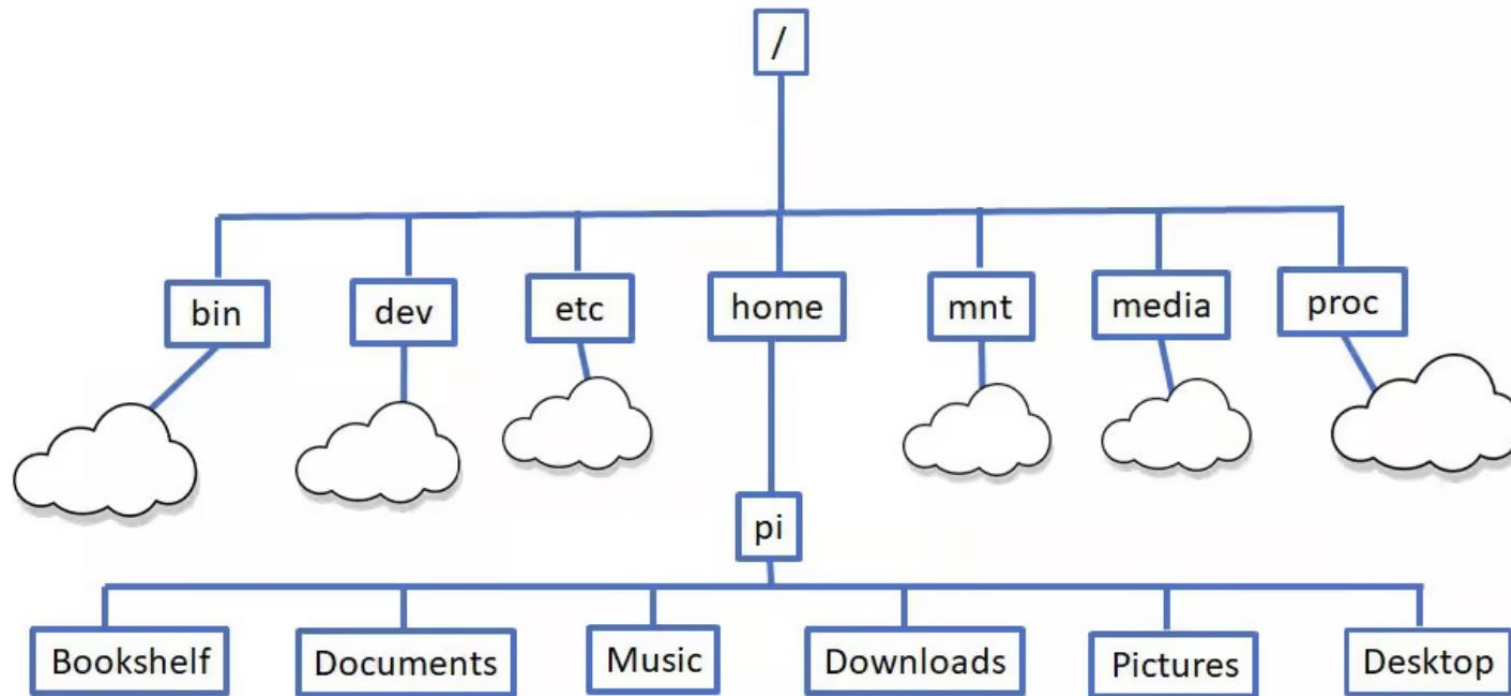
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The Linux Terminal





Explore files and folders

Command	Description
pwd	Show the current directory you are in
ls	List files in the current directory; add -l or -a for more detail or to see hidden files
cd /path/goes/here	Change directory (for example cd /home/pi/Documents)
mkdir myfolder	Create a new directory named myfolder
cp file1 file2	Copy a file; cp -r dir1 dir2 copies a whole directory
mv old new	Move or rename files/directories
rm file	Delete a file; use rm -r folder carefully to remove a directory



Helpful navigation and history tips

Command or Keyboard Entry	Description
clear	Clear the terminal screen.
history	Show a numbered list of past commands; run one again with !number.
Up/Down arrow keys	Scroll through previously used commands to run or edit them.



View and edit files

Command	Description
nano file.txt	Open a simple text editor in the terminal to create or edit a file
less file.txt	View a long file one page at a time (use q to quit)
cat file.txt	Print the contents of a text file to the screen



System information and status

Command	Description
hostname -I	Show the Pi's IP address
uname -a	Show Linux kernel and system details
df -h	Show disk space usage in human-readable form
free -h	Show memory (RAM) usage
uptime	See how long the Pi has been running
top or htop	Monitor running processes and CPU usage (press q to exit top).



IP address of your RPi connected to the network using WiFi

ifconfig -a

```
pi@raspberrypi017: ~  
File Edit Tabs Help  
pi@raspberrypi017:~$ ifconfig -a  
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255  
    ether ea:e0:b9:19:85:8f txqueuelen 0 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 169.254.247.201 netmask 255.255.0.0 broadcast 169.254.255.255  
    inet6 fe80::21e4:d41c:791a:a9d7 prefixlen 64 scopeid 0x20<link>  
    ether dc:a6:32:98:38:24 txqueuelen 1000 (Ethernet)  
    RX packets 1874 bytes 166519 (162.6 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 1723 bytes 1258105 (1.1 MiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 53 bytes 5297 (5.1 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 53 bytes 5297 (5.1 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 10.198.24.23 netmask 255.255.252.0 broadcast 10.198.27.255  
    inet6 fe80::3d8a:ccb:6a3a:2f0 prefixlen 64 scopeid 0x20<link>  
    ether dc:a6:32:98:38:25 txqueuelen 1000 (Ethernet)  
    RX packets 12 bytes 1379 (1.3 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 67 bytes 8066 (7.8 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
pi@raspberrypi017:~$
```



Building up the IoT Architecture and Ecosystem

Icon Sources:

sensor by Carolina Cani; sensor by Pham Duy Phuong Hung, sensor by Tippawan Sookruay, sensor by Lorenzo:

<https://thenounproject.com/browse/icons/term/sensor/>

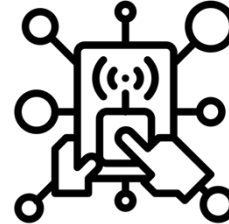
fire sensor by LAFS: <https://thenounproject.com/browse/icons/term/fire-sensor/>

Ultrasound by Shocho: <https://thenounproject.com/browse/icons/term/ultrasound/>

Network by Solikin; Network by Tippawan: <https://thenounproject.com/browse/icons/term/network/>

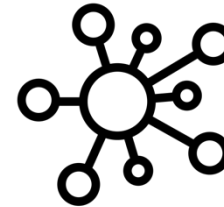
application by Chaowalit Koetchuea: <https://thenounproject.com/browse/icons/term/application/>

Information-layer



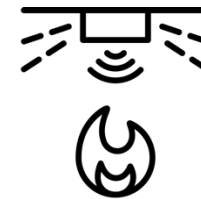
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Communication-layer



Connectivity

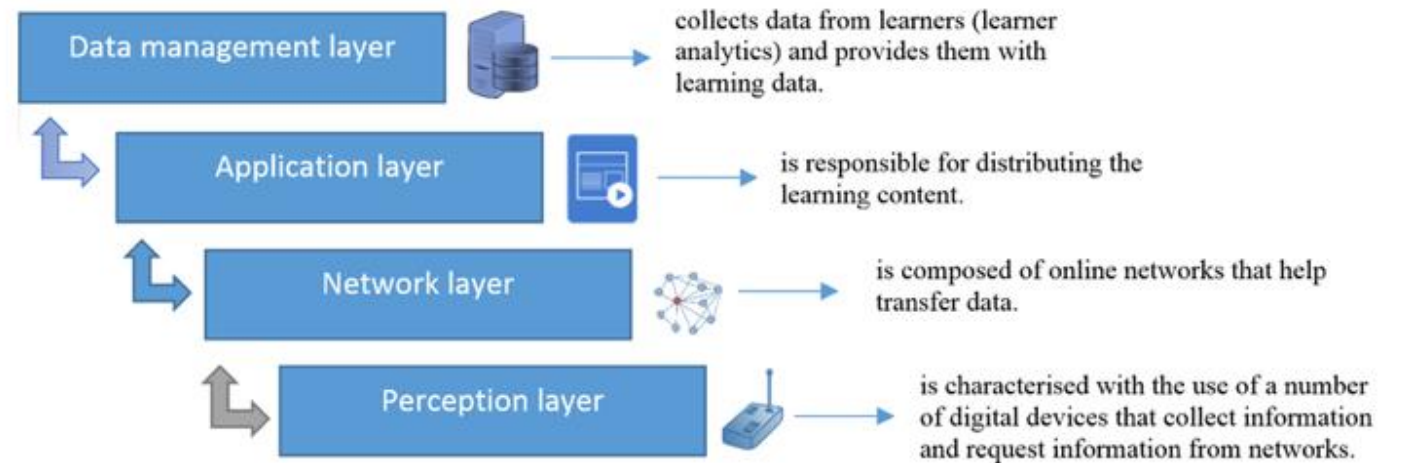
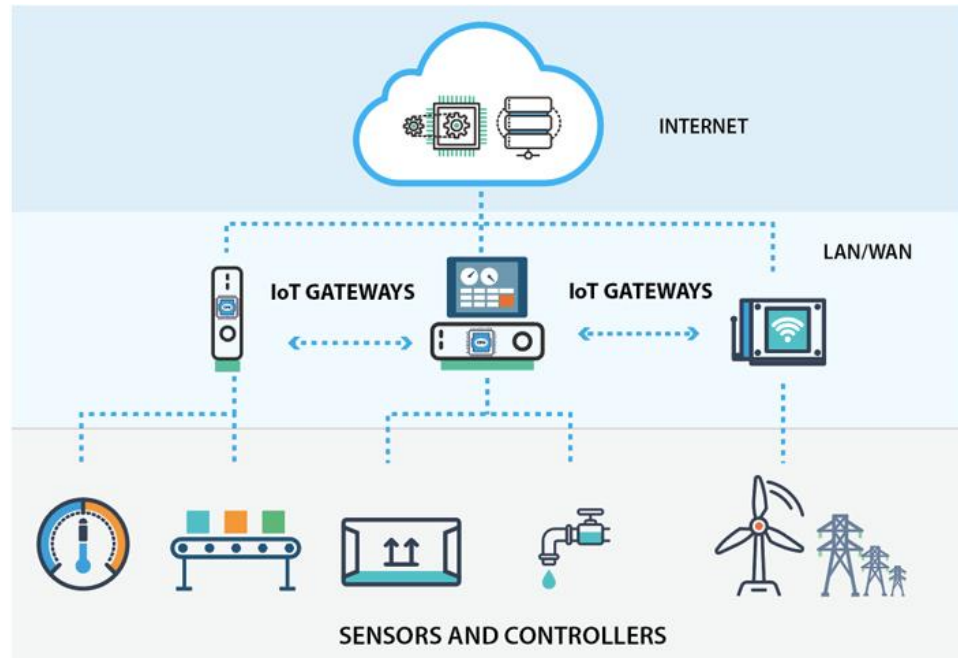
Sensor-layer



Things



Refining and defining the 3-Layer IoT Architecture



The 3-Layer IoT Architecture

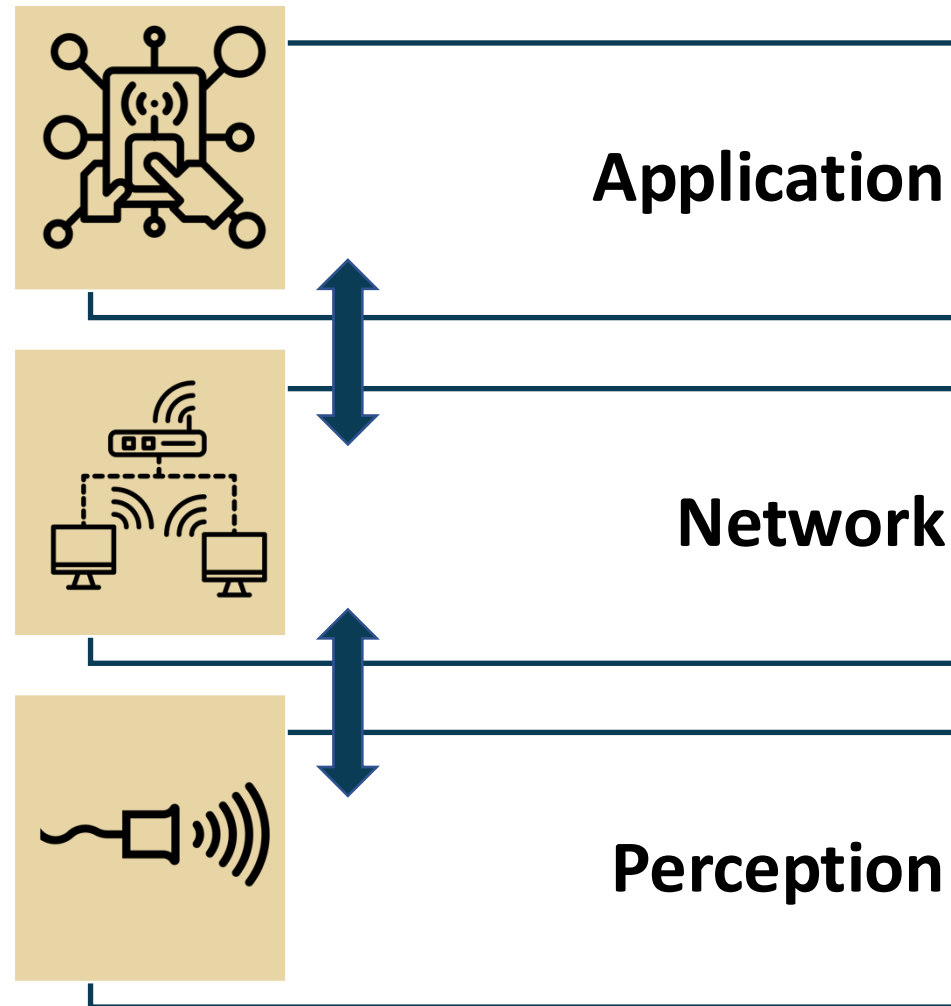
Icon Sources:

sensor by Carolina Cani:, sensor by Pham Duy Phuong Hung, sensor by Tippawan Sookruay, sensor by Lorenzo:

<https://thenounproject.com/browse/icons/term/sensor>

wifi network by Matthias Hartmann:: <https://thenounproject.com/browse/icons/term/wifi-network/>

application by Chaowalit Koetchuea: <https://thenounproject.com/browse/icons/term/application/>



Skeleton of the updated Python program written for Raspberry Pi

```
import library1 as name1  
import RPi.GPIO as GPIO  
import time
```

Import libraries that are relevant for interaction with the Raspberry Pi hardware such as GPIO pins, camera ports etc.

```
GPIO.setmode(GPIO.BOARD)
```

```
GPIO.setup(12, GPIO.OUT)
```

Set up GPIO pins as data outputs or inputs for sensors and actuators

```
def function_name(arguments):  
    statement  
    statement  
    ...  
    return value
```

Create user-defined functions to modularize your code and make it easy to work.

Create user-defined functions to release the GPIO pins

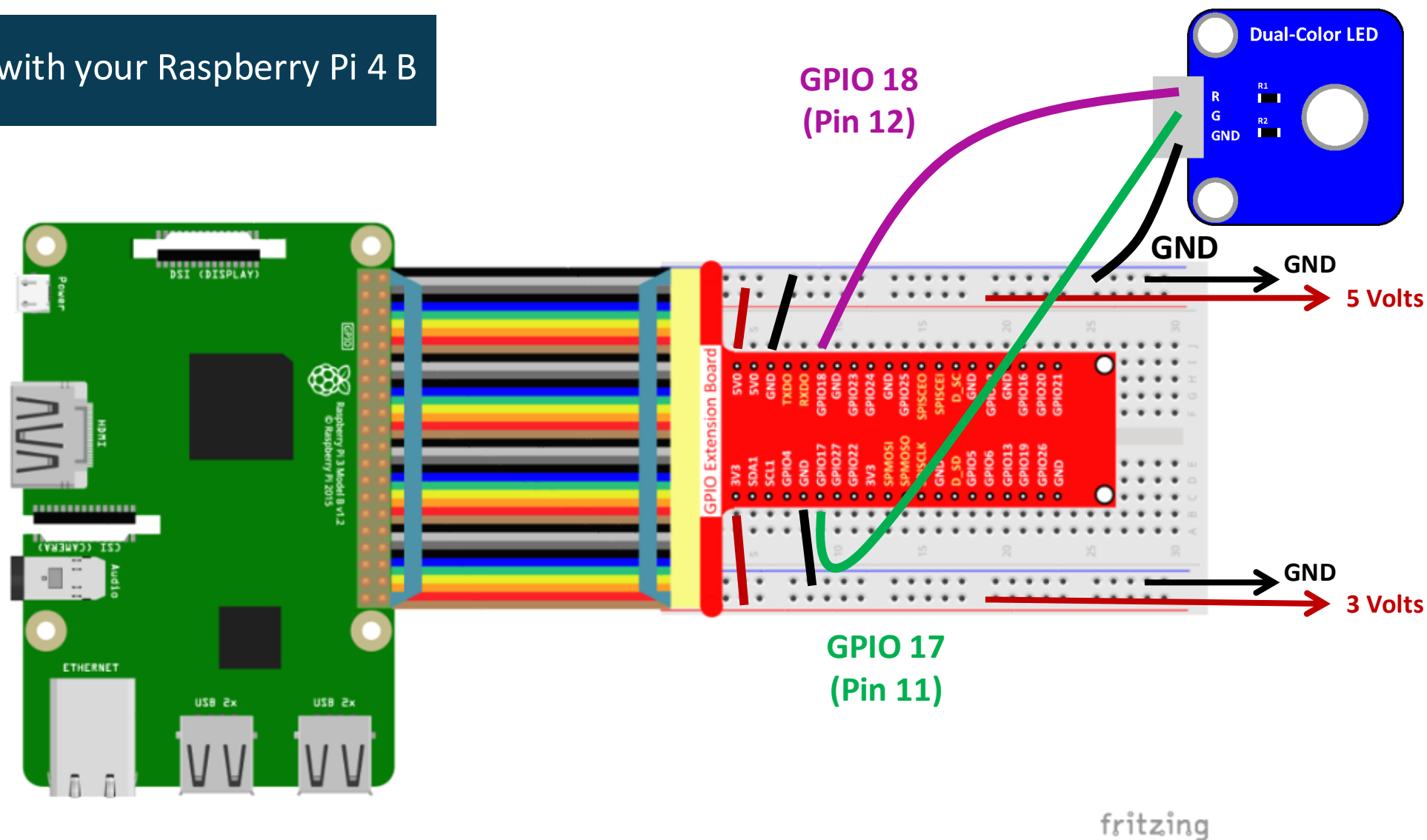
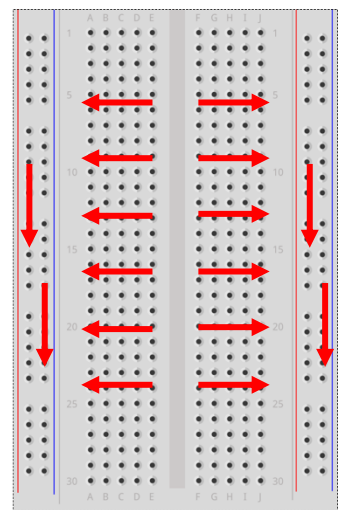
```
if __name__ == "__main__":  
    try:  
        function_name1()  
    except KeyboardInterrupt:  
        function_name2()
```

Create entry point into the program and pull all functions in

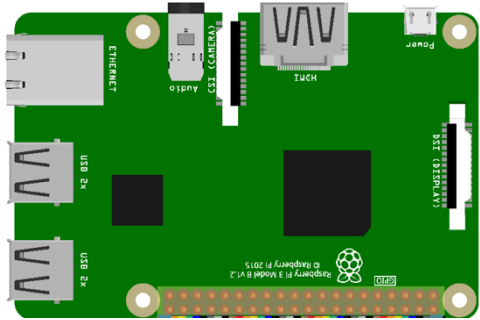
Create a keyboard-interrupt exception clause



Light up an LED with your Raspberry Pi 4 B



Another python
code to kick start
your Raspberry Pi
Model 3 B+ (RPi)



```
import RPi.GPIO as GPIO
import time
```

```
GPIO.setmode(GPIO.BOARD)
```

GPIO Extension Board			
1	• 3V3	• 5V0	2
3	• SDA1	• 5V0	4
5	• SCL1	• GND	6
7	• GPIO4	• TXD0	8
9	• GND	• RXD0	10
11	• GPIO17	• GPIO18	12
13	• GPIO27	• GND	14
15	• GPIO22	• GPIO23	16
17	• 3V3	• GPIO24	18
19	• SPMOSI	• GND	20
21	• SPMOSO	• GPIO25	22
23	• SPISCLK	• SPISCEO	24
25	• GND	• SPISCEI	26
27	• D_SD	• D_SC	28
29	• GPIO5	• GND	30
31	• GPIO6	• GPIO12	32
33	• GPIO13	• GND	34
35	• GPIO19	• GPIO16	36
37	• GPIO26	• GPIO20	38
39	• GND	• GPIO21	40

```
GPIO.setup(12, GPIO.OUT)
```

```
def loop():
    while True:
        GPIO.output(12, GPIO.HIGH)
        time.sleep(0.5)
        GPIO.output(12, GPIO.LOW)
        time.sleep(0.5)
```

```
def destroy():
    GPIO.output(12, GPIO.LOW)
    # Turn off all leds
    GPIO.cleanup()
```

```
if __name__ == "__main__":
    try:
        loop()
    except KeyboardInterrupt:
        destroy()
```



Setting up a Python virtual environment - “Sandbox” your Python projects

“Unwearyingly working and making changes to the global Python environment can break Python-based system-level tools, and remedying the situation can become a major headache.”

Gary Smart, Practical Python Programming for IoT

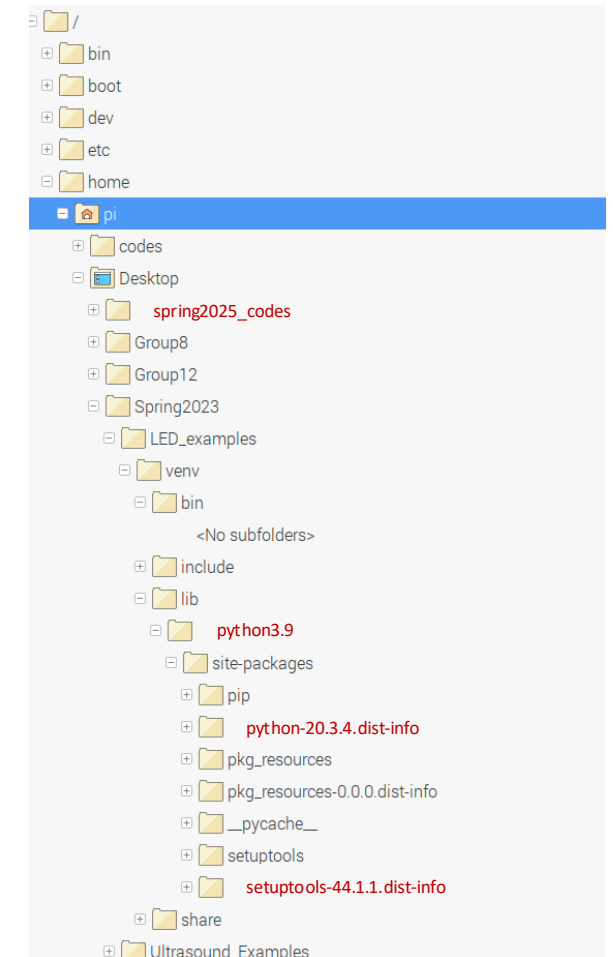


Create python virtual environments

- To “sandbox” our Python projects
- Avoid adverse interference with
 - system-level Python utilities
 - or other Python projects.

```
pi@raspberrypi: ~/Desktop/spring2025_codes/LED_examples
File Edit Tabs Help
pi@raspberrypi:~ $ cd Desktop/spring2025_codes/LED_examples/
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ python -m venv venv
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ source venv/bin/activate
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ which python
/home/pi/Desktop/spring2025_codes/LED_examples/venv/bin/python
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ python --version
Python 3.9.2
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ deactivate
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
```

Anatomy of the python virtual environments



Activate python-venv & Install pip

```
pi@raspberrypi: ~/Desktop/spring2025_codes/LED_examples
File Edit Tabs Help

pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ source venv/bin/activate
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ pip install --upgrade pip
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple
Requirement already satisfied: pip in ./venv/lib/python3.9/site-packages (20.3.4)
Collecting pip
  Downloading https://www.piwheels.org/simple/pip/pip-25.0-py3-none-any.whl (1.8 MB)
    |██████████| 1.8 MB 420 kB/s
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 20.3.4
    Uninstalling pip-20.3.4:
      Successfully uninstalled pip-20.3.4
  Successfully installed pip-25.0
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ pip list
Package Version
-----
pip      25.0
pkg_resources 0.0.0
setuptools 44.1.1
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
```

Review of the code provided to you

```
Thonny - /home/pi/Desktop/spring2025_co...LED_examples/gpio_pkg_check.py @ 32:75
File Edit View Run Tools Help

gpio_pkg_check.py x
1 """
2 File: chapter01/gpio_pkg_check.py
3 Source: Practical Python Programming for IoT
4 Author: Gary Smart
5
6 This Python script checks for the availability of various Python GPIO Library Packages f
7 It does this by attempting to import the Python package. If the package import is succes
8 we report the package as Available, and if the import (or import initialization) fails f
9 we report the package as Unavailable.
10
11 Dependencies:
12 pip3 install gpiozero pigpio
13
14 Built and tested with Python 3.7 on Raspberry Pi 4 Model B
15 """
16
17 try:
18     import gpiozero
19     print('GPIOZero Available')
20 except:
21     print('GPIOZero Unavailable. Install with "pip install gpiozero"')
22
23 try:
24     import pigpio
25     print('PiGPIO Available')
26 except:
27     print('PiGPIO Unavailable. Install with "pip install pigpio"')
28
29 try:
30     import RPi.GPIO
31     print('RPi.GPIO Available')
32 except:
33     print('RPi.GPIO Unavailable. Install with "pip install RPi.GPIO"')
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```

Install GPIO packages using the code provided.

Note: You are still in python-venv

Checks and balances:
pip list
pip freeze > requirements

```
pi@raspberrypi: ~/Desktop/spring2025_codes/LED_examples
File Edit Tabs Help
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ python3 pkg_check.py
GPIOZero Unavailable. Install with "pip install gpiozero"
PiGPIO Unavailable. Install with "pip install pigpio"
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ pip install gpiozero RPi.GPIO
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple
Collecting gpiozero
  Downloading https://www.piwheels.org/simple/gpiozero/gpiozero-2.0.1-py3-none-any.whl (150 kB)
Collecting pigpio
  Downloading https://www.piwheels.org/simple/pigpio/pigpio-1.78-py2.py3-none-any.whl (39 kB)
Collecting RPi.GPIO
  Downloading https://www.piwheels.org/simple/rpi-gpio/RPi.GPIO-0.7.1-cp39-cp39-linux_armv7l.whl (66 kB)
Collecting colorzero (from gpiozero)
  Downloading https://www.piwheels.org/simple/colorzero/colorzero-2.0-py2.py3-none-any.whl (26 kB)
Collecting importlib-resources~=5.0 (from gpiozero)
  Downloading https://www.piwheels.org/simple/importlib-resources/importlib_resources-5.13.0-py3-none-any.whl (32 kB)
Collecting importlib-metadata~=4.6 (from gpiozero)
  Downloading https://www.piwheels.org/simple/importlib-metadata/importlib_metadata-4.13.0-py3-none-any.whl (23 kB)
Collecting zipp>=0.5 (from importlib-metadata~=4.6->gpiozero)
  Downloading https://www.piwheels.org/simple/zipp/zipp-3.21.0-py3-none-any.whl (9.6 kB)
Requirement already satisfied: setuptools in ./venv/lib/python3.9/site-packages (from colorzero->gpiozero) (44.1.1)
Installing collected packages: RPi.GPIO, pigpio, zipp, colorzero, importlib-resources, importlib-metadata, gpiozero
Successfully installed RPi.GPIO-0.7.1 colorzero-2.0 gpiozero-2.0.1 importlib-metadata-4.13.0 importlib-resources-5.13.0 pigpio-1.78 zipp-3.21.0
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
```

```
pi@raspberrypi: ~/Desktop/spring2025_codes/LED_examples
File Edit Tabs Help
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ pip list
Package Version
-----
colorzero 2.0
gpiozero 2.0.1
importlib-metadata 4.13.0
importlib-resources 5.13.0
pigpio 1.78
pip 25.0
pkg_resources 0.0.0
RPi.GPIO 0.7.1
setuptools 44.1.1
zipp 3.21.0
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ python3 pkg_check.py
GPIOZero Available
PiGPIO Available
RPi.GPIO Available
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ pip freeze > requirements
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
```

pip stands for “Python installs packages”.

pip queries the Python Package Index, or simply PyPi

Web source for PyPi: <https://pypi.org>



Run the python codes
Blink your LEDs
Note: You are still in python-venv

deactivate your venv

```
pi@raspberrypi: ~/Desktop/spring2025_codes/LED_examples
File Edit Tabs Help
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ which python
/home/pi/Desktop/spring2025_codes/LED_examples/venv/bin/python
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ python --version
Python 3.9.2
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ python gpio.py check.py
GPIOZero Available
PiGPIO Available
RPi.GPIO Available
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ python LED.py
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(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ clear
```

```
pi@raspberrypi: ~/Desktop/spring2025_codes/LED_examples
File Edit Tabs Help
(venv) pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ deactivate
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
```



Running a python script at boot cron, the UNIX scheduler

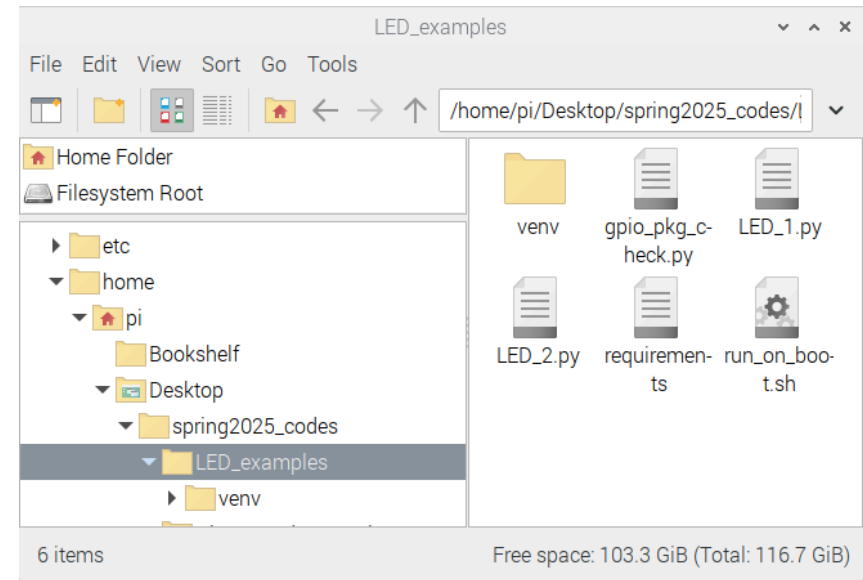
Awesome IoT project will run automatically every time you start your Raspberry Pi



Review run_on_boot.sh

Note: You are no longer in python-venv

Let's make sure we have a file-folder structure that looks some what like this



Let's open the *run_on_boot.sh* on a terminal to review the script



Absolute paths to

1. Python interpreter in venv
2. Python script
3. Locate the LOG-file



The last two lines tie everything together

1. echo [updates LOG-file with string in quotes]
2. Next line runs py-file and logs output on boot
3. 2>&1 [logs any additional errors during run]



```
pi@raspberrypi: ~/Desktop/spring2025_codes/LED_examples
File Edit Tabs Help
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ cat run_on_boot.sh
#!/bin/bash
#File: LED_examples/run_on_boot.sh

# Absolute path to Virtual Environment python interpreter
PYTHON=/home/pi/Desktop/spring2025_codes/LED_examples/venv/bin/python

# Absolute path to Python script
SCRIPT=/home/pi/Desktop/spring2025_codes/LED_examples/LED_1.py

# Absolute path to output log file
LOG=/home/pi/Desktop/spring2025_codes/LED_examples/LED_1.log

echo -e "\n##### STARTUP $(date) #####\n" >> $LOG

$PYTHON $SCRIPT >> $LOG 2>&1
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
```



chmod u+x run_on_boot.sh
*current **U**ser, make the file e**X**ecutable*

```
pi@raspberrypi: ~/Desktop/spring2025_codes/LED_examples
File Edit Tabs Help
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ cat run_on_boot.sh
#!/bin/bash
#File: LED_examples/run_on_boot.sh

# Absolute path to Virtual Environment python interpreter
PYTHON=/home/pi/Desktop/spring2025_codes/LED_examples/venv/bin/python

# Absolute path to Python script
SCRIPT=/home/pi/Desktop/spring2025_codes/LED_examples/LED_1.py

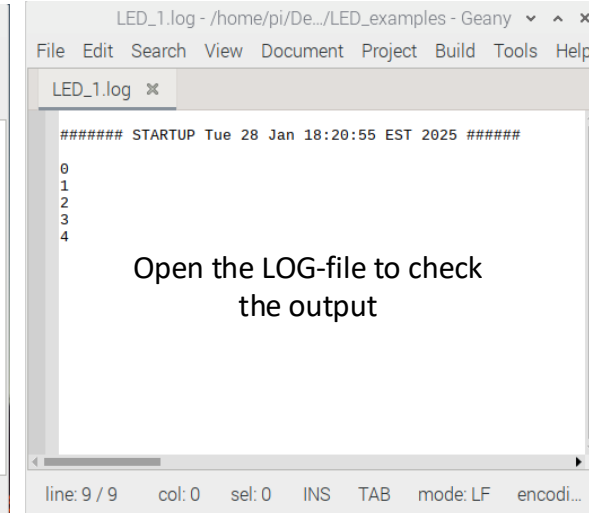
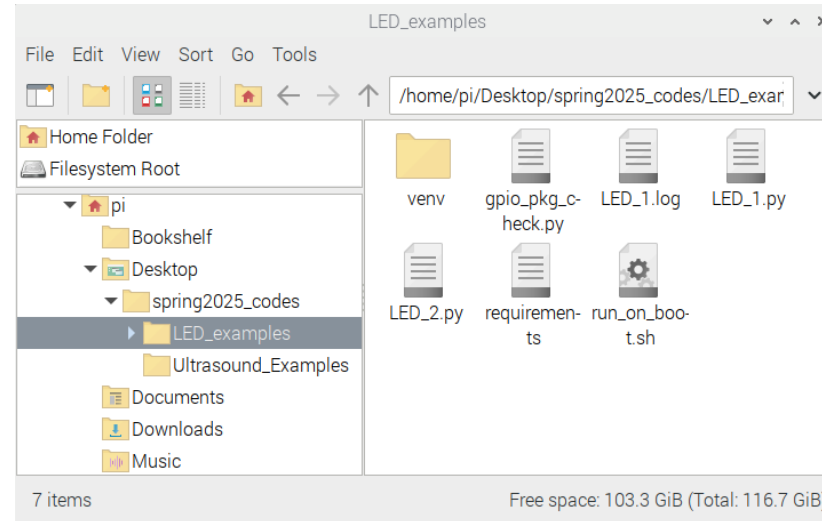
# Absolute path to output log file
LOG=/home/pi/Desktop/spring2025_codes/LED_examples/LED_1.log

echo -e "\n##### STARTUP $(date) #####\n" >> $LOG

$PYTHON $SCRIPT >> $LOG 2>&1
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ chmod u+x run_on_boot.sh
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
```

./run_on_boot.sh

```
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ ./run_on_boot.sh
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $
```



crontab -e

This is a list of editors used in terminal with Linux or Unix OS

I recommend you choose “nano” or the default Option-1 and hit “Enter”.

“vim” and “ed” are also popular with advanced or legacy users

“code” will open up VS Code that slows down the Raspberry Pi OS

```
$ crontab -e
```

*no crontab for pi - using an empty one
Select an editor. To change later, run 'select-editor'.*

- -- --> {
1. /bin/nano <---- easiest
 2. /usr/bin/vim.tiny
 3. /usr/bin/code
 4. /bin/ed

Choose 1-4 [1]: 1



crontab -e

\$ crontab -e

@reboot /home/pi/Desktop/spring2026_codes/LED_examples/run_on_boot.sh &

"Do not forget the & character at the end of the line. This makes sure the script runs in the background."

Gary Smart, Practical Python Programming for IoT

```
pi@raspberrypi: ~/Desktop/spring2025_codes/LED_examples
File Edit Tabs Help
GNU nano 5.4 /tmp/crontab.MInc7f/crontab *
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
@reboot /home/pi/Desktop/spring2026_codes/LED_examples/run_on_boot.sh &
```

Try manually ./run_on_boot.sh
then
sudo reboot

```
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ ./run_on_boot.sh
pi@raspberrypi:~/Desktop/spring2025_codes/LED_examples $ sudo reboot
```

To stop a cron job you can type this on the terminal:
sudo service cron stop



Someone should summarize what we learned today

